

Edward (Ted) Ivanac

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734-890-7100

EDUCATION	University of Michigan , Ann Arbor, MI	August 2025 - May 2027
	<i>Master of Science (MS)</i> , Robotics	GPA: 4.00
	• Notable Courses: Computational Human-Robot Interaction • Awards: Roger and Carin Levine Ehrenburg Fellowship	
	University of Michigan , Ann Arbor, MI	August 2019 - May 2024
	<i>Bachelor of Science in Engineering (BSE)</i> , Robotics	GPA: 3.41
	<i>Bachelor of Science in Engineering (BSE)</i> , Electrical Engineering	
	• Notable Courses: Simultaneous Localization & Mapping, Robot Operating Systems, Autonomous Vehicles • Awards: University Honors, Dean's List (May 2023, December 2023, May 2024) Michigander Scholar's Award	
TEACHING	Duderstadt Center Fellowship , Ann Arbor, MI	
EXPERIENCE	<i>Lead Fabrication Fellow</i>	September 2020 - May 2024
	• Instituted an interdisciplinary fellowship program and training material for users in both multimedia and fabrication spaces • Developed and facilitated 4 workshops multiple times over three years on topics including Arduino, Raspberry Pi, Soldering, and CAD Design • Authored instructional guides for students on topics including Additive Manufacturing, CAD Design, and PCB Design	
	ENGR 110 , Ann Arbor, MI	
	<i>Instructional Aide</i>	August 2023 - December 2024
	• Facilitated weekly discussion sections for ENGR 110 class, fostering an engaging learning environment for students in the Electrical and Robotics Engineering departments • Contributed to the refinement of existing course material through weekly feedback sessions on experiences with students • Effectively communicated the values and distinctive features of the Robotics department to peers, fostering a sense of community and enthusiasm for the field	
	Michigan Engineering Zone , Detroit, MI	
	<i>Lead Robotics Mentor Coordinator</i>	September 2020 - May 2022
	• Created an educational robotics kit and curriculum that taught C programming as well as using encoders in a control loop through Arduino robotics hardware • Led a 20-person mentor group to create technical and non-technical workshops for high school students in the FIRST Robotics Competition	
INDUSTRY	General Motors , Warren, MI	
EXPERIENCE	<i>Assistant Vehicle Systems Engineer</i>	July 2024 - August 2025
	• Develops Corporate Common Components plans that ensure maximum use across vehicle programs and architectures. Ensure each component is integrated and at the right cost/risk profile • Drives completion of associated business case impacts by working with interdisciplinary teams of specialists in tech, manufacturing, materials, and design along with potential buyers and PPCO	

Ford Motor Company, Dearborn, MI

ADAS Product Development Intern

May 2022 - July 2022

- Developed four ADAS data analysis tools in MATLAB to post-process vehicle features success metrics specifically for the Lane Change Assist feature, surpassing target goal by three
- Implemented and verified pre-made rapid-install device with a built-in GPS and IMU to track vehicle dynamics data on autonomous vehicles, resulting in more efficient methods of instrumenting vehicles

PROJECTS

Brushless DC Motor Driver

Robot Sensors & Signals Course

April 2024

- Codeveloped brushless motor driver for use in the MBot platform with a team of three other robotics engineering students
- Redesigned open-source SimpleFOC Arduino shield as a PCB to be compatible with Raspberry Pi Pico microcontroller
- Designed test fixtures in Fusion 360 for PCB board and gimbal brushless motor to test torque control of the motor using an absolute magnetic encoder

Forward/Inverse Kinematics

Robot Operating Systems Course

January 2024 - April 2024

- Implemented forward kinematics in Javascript to map joint/link configurations to end effector poses URDF-style Javascript robot simulations
- Implemented inverse kinematics in Javascript to infer joint states to reach a desired end effector pose
- Developed Finite State Machine to cycle robot description through a series of end effector poses to create a robot dance routine

MBot SLAM & Navigation

SLAM & Navigation Course

September 2023 - December 2023

- Developed a program in C++ that generates a map in ROS2 from live LIDAR data
- Integrated a particle filter algorithm for real-time localization inside of an environment for simultaneous localization and mapping
- Implemented A* path planning algorithms to navigate a mobile robot through a maze autonomously

Autonomous Vehicle Simulation

Autonomous Vehicles Course

September 2023 - December 2023

- Implemented a Kinematic Bicycle model in MATLAB with realistic force modeling including aerodynamic drag and tire rolling resistance
- Developed and tuned PI controllers for velocity control and steering wheel control to autonomously navigate the vehicle along a series of predetermined waypoints
- Implemented A* path planner library into a simulation model to plan a path and navigate through a map with randomly generated obstacles

TECHNICAL SKILLS

Software : C++, MATLAB, Python, Github, Simulink, Javascript, Simulink, Fusion 360, EAGLE CAD

Hardware : PCB Design, 3D Printing, Raspberry Pi Pico, Lidar, CAN, IMUs